[INCH-POUND]
A-A-51179A
July 17, 1996
SUPERSEDING
A-A-51179
16 April 1990

COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINES, UTILITY, BENCH AND PEDESTAL

The General Services Administration has authorized the use of this commercial item description as a replacement for A-A-51179 for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers the requirements for electric motor driven, two-wheel, utility type, bench and pedestal mounted, grinding machines.

1.2 Intended use. The grinding machines covered by this CID are intended for grinding of tools and workparts which are held in the operator's hands. It does not cover cutting-off, cylindrical, internal, portable, saw gumming, snagging or tool grinding machines.

2. CLASSIFICATION

2.1 Classification. The grinding machines shall be of the following styles and sizes. The style and size to be furnished shall be as specified (see 7.2).

2.2 Style. Style I - Bench Mounted
Style II - Pedestal Mounted

2.3 Size. Size 6 - 6" grinding wheel
Size 7 - 7" grinding wheel
Size 8 - 8" grinding wheel
Size 10 - 10" grinding wheel
Size 12 - 12" grinding wheel
Size 14 - 14" grinding wheel

Beneficial comments, recommendations, additions, deletions, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS

3.1 Requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement or repair. The grinding machine shall meet the requirements of American National Standards Institute (ANSI) B11.9 or shall be listed with Underwriter's Laboratories (UL) as meeting the requirements of UL 987. The grinding wheels and associated mechanisms shall meet the requirements of ANSI B74.2 and ANSI B7.1. The machine shall have a grinding wheel mounted on each end of a spindle and shall meet the performance requirements shown in Table I for the machine size indicated. Unless otherwise specified (see 7.2), the machine shall include one coarse (36 grit) and one medium (60 grit) wheel. Unless otherwise specified (see 7.2), the height and wheel-to-wheel centerline dimensions shall be in accordance with the manufacturer's standard practices.

3.2 Electrical requirements. The machine shall operate from a dual 230/460 volt, 3 phase, 60 Hz circuit. Unless otherwise specified (see 7.2), the machine shall be wired for 230 volt operation. The motor shall be energy efficient and shall conform to NEMA MG-1.

TABLE I. Performance Requirements.

Machine Size (Grinding Wheel Diameter)	Spindle Speed (RPM)	Spindle Motor Rating (horsepower)
Size 6	3450	1/3
Size 7	3450	1/2
Size 8	3450	3/4
Size 10	1725	1
Size 12	1800	2
Size 14	1800	3

NOTE: The spindle speed and motor rating shall not be less than the values shown for the machine size indicated.

3.3 Drill grinding attachment. When specified (see 7.2), a drill grinding attachment shall be furnished with all necessary equipment for accurately grinding drill point angles from 90° to 130° on twist drills or flat drills having straight or taper shanks.

3.4 Wheel dresser. When specified (see 7.2), a wheel dresser of the revolving cutter type shall be supplied.

4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirement, and promotes economically advantageous life cycle costs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be met during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING

6.1 Packaging requirements. Unless otherwise specified (see 7.2), the item shall be packaged in accordance with the manufacturer's standard practices.

7. NOTES

7.1 Referenced documents.

7.1.1 Government documents. Copies of Military and Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this Commercial Item Description may be obtained from the following addresses:

a. Underwriters Laboratories (UL), Inc., 333 Pfingsten Road, Northbrook, IL 60062.
b. American National Standards Institute, 1430 Broadway, New York, NY 10018.
c. National Electrical Manufacturers Association, 2101 L St. N.W., Washington, DC. 20037.

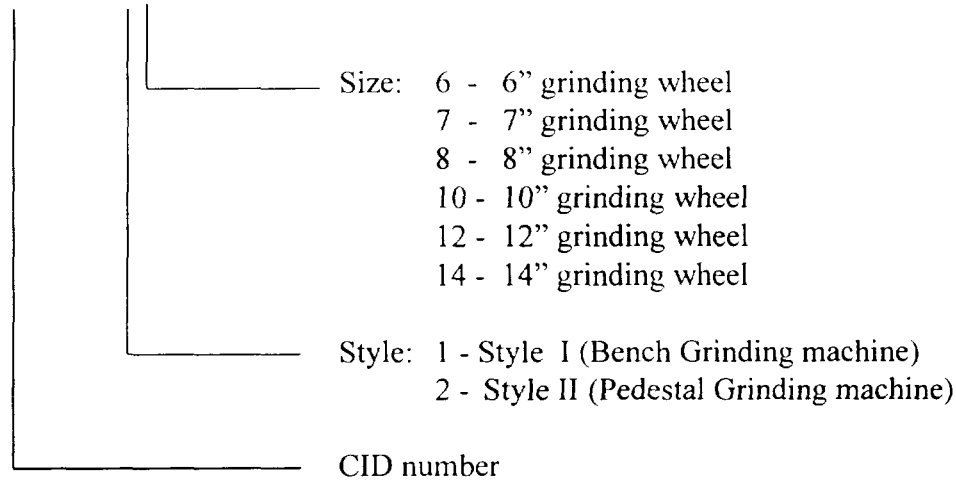
7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Style and size of grinding machine required (see 2.0).
c. Grinding wheels, if different (see 3.1).
d. Grinding machine height, if different (see 3.1).
e. Grinding machine wheel-to-wheel centerline, if different (see 3.1).
f. Electric power requirements, if different (see 3.2).
g. Drill grinding attachment, if required (see 3.3).
h. Wheel dresser, if required (see 3.4).
i. Packaging, if different (see 6.1).

Source: https://assist.dla.mil --Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

AA51179A-XX



MILITARY INTERESTS:

Custodians
Army - AL
Air Force - 99

Review Activities
Air Force -84
Navy - SH

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

PREPARING ACTIVITY:
DLA-GS

(Project 3415-0209)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:52Z
Check the source to verify that this is the current version before use.